



Homework 7
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SVM
(Support Vector Machines)



Motivation

In the upcoming lecture we will cover Support Vector Machine (SVM), a very powerful classification/ regression method, with a sound mathematical basis. In order to understand the math used, you will review the method of Lagrange multipliers for Optimization and gradient descent.

We will use both methods during class.



Part 1 – Method of Lagrangian Multipliers and Duality

Estimated time: 45min to watch the videos + 25 min for the questions

The Lagrangian Duality and KKT should already be known to you from MVI, if you have taken MVI, in which case this part is more of a review for you. If you have not taken MVI yet, this may be new material (which you will later review in MVI!).

- ♦ **Watch the following videos**

- 1) What Is Mathematical Optimization?

<https://www.youtube.com/watch?v=AM6BY4btj-M>

- 2) Convexity and The Principle of Duality

<https://www.youtube.com/watch?v=d0CF3d5aEGc>

- 3) The Karush–Kuhn–Tucker (KKT) Conditions and the Interior Point Method for Convex Optimization

<https://www.youtube.com/watch?v=uh1Dk68cfWs>



Part 2 – (Stochastic) Gradient Descent

Estimated time: 25min to watch the videos + 25 min for the questions

There are many ways to find the optimal (minimal or maximal) value for a (complex) function – gradient descent is one of the most basic ones. In practice, more advanced methods are usually employed, but it is important to understand the basic idea of gradient descent. The following video by StatQuest will introduce you to gradient descent incl. stochastic gradient descent (Josh Stamer from StatQuest is another youtuber, who explains very well and very visually – we will watch more videos from him in the upcoming classes/homeworks). This video “requires” that you are already familiar with linear regression and the least squares method – do not worry about this, you do not really need to know either. For linear regression, all you need to know is that it tried to find best “best fitting” line to a set of points (in his example, three points in two dimensions). We will cover linear regression in depth in later classes. Regarding least squares, we will also cover this later in class, all he does in this video is to compare the results of these two optimization methods, so no need to understand least squares just now.

- ♦ **Watch the following video**

- 1) Gradient Descent, Step-by-Step

- <https://www.youtube.com/watch?v=sDv4f4s2SB8>

- ♦ **Please answer the following questions**

- 1) What do you think will happen if the learning rate is too large? Try to explain this visually. param would approach 0 too quickly to find optimal
 - 2) His example has a very simple loss function (for intercept and slope) which is convex and has one global minimum. Imagine a much more complex loss function with many (local) minima – what do you think might happen in this case, if the learning rate is too low, and what might happen if the learning rate is too large?

if learning rate too low, then it might not find optimals closer to 0 before reaching max iteration count

if learning rate too large, then it will approach 0 too quickly and not find all minima

- ♦ **Compare and discuss your answers with one other student in the class.**

- If your answers are different, try to convince each other that yours are correct!



Congratulations!

